Ayasha Hossain Jui

Software Engineer

+880 1521-410529 | ayashajui1110@gmail.com | linkedin.com/in/ayasha-hossain-jui | github.com/ayashaJui | ayashaJui.github.io

Summary

Software Engineer with 3+ years of experience designing and delivering scalable, production-grade backend systems in the FinTech domain. Proficient in Java, Spring Boot, Spring Security, and Spring Data JPA (Hibernate), with hands-on experience building concurrency-safe transaction workflows (optimistic locking, atomic and idempotent balance updates) and event-driven systems for real-time market data using Kafka and RabbitMQ. Experienced with PostgreSQL, SQL Server, MySQL, and MongoDB, as well as Docker, CI/CD, and JWT/OAuth2 authentication. Ownership-driven, with a strong focus on clean code, sound architectural decisions, and cross-team knowledge sharing.

Experience

Software Engineer March 2023 – Present

Dohatec New Media – Independent Software Vendor *Dhaka, Bangladesh*

- Develop and maintain production microservices in **Java Spring Boot** for a live securities trading platform; set up the service registry, config server, and API gateway for a scalable, maintainable architecture.
- Built the **financial service** for PQCal Global, handling all balance credits and debits with **atomic, idempotent, and concurrency-safe** updates using database transactions and **optimistic locking** (JPA `@Version` column) to prevent duplicate or conflicting concurrent requests.
- Built the **market data service** for PQCal Global, consuming real-time market data streams from **Kafka** and exposing **RESTful APIs** that serve the data to the Angular UI.
- Developed versioned **RESTful APIs** for user management, authentication, financial services, and market analysis modules; followed clean code standards and took part in code reviews.
- Wrote and optimized queries for **SQL Server** and **PostgreSQL** using **Spring Data JPA / Hibernate**; diagnosed and tuned slow queries and APIs to support high-volume financial data processing.
- Implemented asynchronous, event-driven inter-service communication with **RabbitMQ** and **Kafka**, improving decoupling and reliability.
- Implemented authentication and authorization with **Spring Security** and **JWT/OAuth2**; extended it into a dedicated auth server for international product expansion.
- Developed reusable **Angular** UI components with PrimeNG and secure token-based authentication flows for full-stack delivery.
- Worked with QA and DevOps on **CI/CD pipelines** and **Docker** deployments; contributed to unit and functional test coverage, system design reviews, and Agile sprint planning.

Projects

PQCal (*Live Production*) Dohatec New Media

Java Spring Boot, Angular, PrimeNG, SQL Server, PostgreSQL, RabbitMQ, Docker *pqcal.dohasecurities.com.bd*

Dohatec's first in-house product — a live securities trading platform used in the Bangladeshi market.

- Contributed to the user management, business rule engine, financial services, and market analysis modules.
- Developed RESTful APIs with Spring Boot and Spring Data JPA; worked with SQL Server and PostgreSQL, writing optimized queries for high-volume data.
- Worked on RabbitMQ messaging and the Angular/PrimeNG UI with JWT-based authentication flows.

PQCal Global (*Under Development*) Dohatec New Media

Java Spring Boot, Angular, Kafka, RabbitMQ, SQL Server, PostgreSQL, CME

International expansion of PQCal for markets outside Bangladesh, with live paper trading via CME.

- Built the **financial service** for balance credits and debits, preventing duplicate or conflicting transactions under concurrent load with **optimistic locking** (version column) and atomic database transactions.
- Built the **market data service**: consume real-time data from **Kafka** topics and expose **REST APIs** that power live views in the Angular UI for global users.
- Integrating **CME (Chicago Mercantile Exchange)** data feeds to power real-time paper trading for global users.
- Set up the microservices infrastructure (service registry, config server, API gateway) with **RabbitMQ** for event-driven communication.
- Contribute across the full stack: Angular frontend and Spring Boot backend, including user management, auth server, and business rules.

Dohatec AI (*RAG Application*) Dohatec New Media
Java Spring Boot, Angular, RAG, LLM, Document Embeddings, Semantic Search

Company AI assistant, built by a cross-functional team, that answers questions from users' own documents through a chat interface.

- Contributed to the **RAG pipeline** as part of the team: document upload and ingestion, text chunking, embedding generation, and indexing for semantic retrieval.
- Worked on the chat flow, where each query retrieves the most relevant document context and passes it to an **LLM** to produce answers grounded in the uploaded documents.
- Developed features in the **Spring Boot** microservice backend and the **Angular** frontend (document management and chat UI), collaborating through code reviews and sprint planning.

Biblioteca (*Personal Project*) github.com/ayashaJui/book_tracking
Java 21, Spring Cloud, OAuth2/JWT, PostgreSQL, Angular, NgRx, Docker

Book-tracking platform (catalogue, reading logs, wishlists, collections, reviews, analytics) built as independent microservices.

- Architected independent **Spring Boot microservices** (user, catalog, user-library) with a database per service, **Eureka** service discovery, **Spring Cloud Config**, and a **Spring Cloud Gateway** as the single entry point.
- Built a dedicated **OAuth2 Authorization Server** issuing **JWT** tokens, with resource servers validating them and the Angular SPA using the authorization code flow.
- Managed schemas with **Flyway**, documented APIs with **OpenAPI/Swagger**, and containerized the core services with **Docker Compose**.
- Built the **Angular 20** frontend with **NgRx** state management, PrimeNG, and Tailwind CSS across 15+ feature modules.

StudyNest (*Live*) studynest-z0r3.onrender.com
Next.js, NestJS, TypeScript, MongoDB, JWT, Swagger, Vitest, GitHub Actions

Interview-prep platform with nested topics, markdown notes, flashcards, quizzes, and a coding-problem tracker.

- Built a full-stack app with a **Next.js** frontend and a separate **NestJS** REST API on **MongoDB**, documented with Swagger and deployed as two services on Render.
- Implemented **SM-2 spaced-repetition** scheduling for flashcards and problem reviews, plus a daily review queue, study streaks, and global search.
- Secured the app with **HttpOnly cookie-based JWT** auth, rate-limited login, and email password reset; added CSV/Markdown import and export.

Ink & Paper inkandpaper-frontend.onrender.com
React, Redux, Node.js, Express.js, MongoDB, Material UI

- Built a full-stack e-commerce bookstore with user authentication, book browsing, author and review exploration, and a shopping cart.
- Designed RESTful APIs with Express and MongoDB; built a responsive React UI using Material UI and Redux for global state.

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, PHP, C++, SQL

Backend: Spring Boot, Spring Cloud (Gateway, Eureka, Config), Spring MVC, Spring Security, Spring Data JPA / Hibernate, Flyway, Node.js, Express.js, NestJS

Frontend: Angular (NgRx, PrimeNG, Tailwind CSS), React (Redux, Material UI), Next.js, HTML, CSS, jQuery

AI & GenAI: Retrieval-Augmented Generation (RAG), LLM Integration, Document Embeddings, Semantic Search, Text Chunking

Databases: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB

Messaging & Streaming: Kafka, RabbitMQ

DevOps & Tools: Docker, Docker Compose, Git, GitHub Actions, CI/CD pipelines, Swagger/OpenAPI, IntelliJ IDEA, Tomcat

Practices: OOP, Design Patterns, Clean Code, JUnit/Mockito, REST API Design & Versioning, Microservices, Event-Driven Architecture, Real-Time Data Processing, Concurrency-Safe Transactions, Optimistic Locking, JWT/OAuth2, API Gateway & Service Discovery, Agile/Scrum, Code Reviews

Education

Khulna University Khulna, Bangladesh
B.Sc. Engineering in Computer Science and Engineering — CGPA: 3.72/4.00 *2018 – 2023*

Publications

Ayasha Hossain Jui, Samia Sharnami, Aminul Islam. *"A CNN Based Approach to Classify Skin Cancers using Transfer Learning"*, 25th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 17–19 December 2022. DOI: 10.1109/ICCIT57492.2022.10055838